E4: Land surface model

palm group

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The aim of this exercise is to get to know the land surface model.

You will study two different cases:

1. You will look at a **short and simple** setup showing the influence of free convection on the soil temperature. Additionally a user defined quantity shall be calculated here for training the handling of user-defined code.
2. You will simulate full **diurnal cycles** and observe night-time dewfall on vegetation.

The following lectures are required:

- Land surface model
- Radiation model
- User defined code

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Part 1 – Setup

Summary of the most relevant namelist settings:

Land surface model

- default 8-layer soil setup
- constant soil temperature profile
- no moisture
- bare soil

Atmospheric boundary layer

- free convection (convective boundary layer without background wind) driven by the net radation at surface
- neutral stratification capped with an inversion at z=800m
- no humidity

Radiation model

- clear-sky around noon
- time step = 20.0 s

Runtime

- 2 hours
- high-frequent output of surface and three-dimensional data

unknown source

Questions:

Look at cross-sections of `theta, w, t_soil, shf, ghf, us*` and `t_surf*` which are found in the in the `_3d` and `_xy` files:

- How do the different variables correlate with one another?
- Down to which depth does the soil feel the boundary layer turbulence?
- What is the influence of a thinner/thicker first soil layer on the reaction of the surface temperature under atmospheric changes?

Output of user-defined 2d cross sections or 3d volume data

For learning how to output user-defined quantities, the absolute value of the horizontal velocity* (`v_h`) shall be calculated in this exercise as an example and output as 2-D cross sections and 3-D volume data.

1. Add the parameter `data_output_user` in the namelist `&user_parameters` in `e4_LSM1_p3d`.
2. Create directory `palm/current_version/JOBS/e4_LSM1/`**`USER_CODE`**
3. Copy the file user_module.f90 (`palm_model_system-v.../packages/palm/model/src/`) to the directory `USER_CODE`
4. Modify the file user_module.f90 as described in https://palm.muk.uni-hannover.de/trac/wiki/doc/app/userint/output

If you want to shorten steps 1 to 4, we provide a modified user code `user_module.f90` along with the steering files of this exercise, where only some question marks (??) need to be replaced!

* Since 2020 this quantity is available as standard output (`wspeed`), but here it should just serve as an example and therefore be calculated again using the user interface.

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Part 1 – Hints

Look up new variables in the PALM documentation!

Be sure that you are aware of what you see and understand how the output is calculated!

palmplot commands

- xy-cross sections:

```
palmplot xy file_1=e4_LSM1_xy.000.nc format_out=pdf \
  file_out=e4_LSM1_plot_xy var='tsurf*_xy ghf*_xy shf*_xy'\
  no_rows=3 no_columns=1 \
  start_time_step=1.0d end_time_step=1.0d
```

ncview

- The high-frequent output in this exercise allows you to have a look at the development of turbulence in the LES. You can loop the cross-sections over the output times in `ncview` and adjust the speed ("Delay" bar).

- Sometimes you might need to adjust the range of the shown output under Range. Leave the mouse cursor over the text box to edit the range. 329.97 Alternatively, klick with right mouse button.
- You can get some decent time series by just clicking somewhere in the output.

The palmplot-commands are just examples and may need to be adjusted to answer the questions! Also the p3d-file may need to be edited to get appropriate outputs!

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Part 2 – Setup

Land surface model

- default 8-layer soil setup
- constant moisture and temperature
- 95% covered with short grass
- no heat capacity of skin-layer

Atmospheric boundary layer

- constant initial profiles of
  - wind (value of geostrophic wind)
  - virtual potential temperature
  - humidity (no clouds/precipitation)

Radiation model

- clear-sky
- start on June 21st at 12:00 UTC

Runtime

- 2.5 days
- output designed for analysis of time series and profiles

Questions:

Can you observe dewfall? (time series: `qsws_liq, xy-section: c_liq`)

- What happens at night? (time series: `ol, z_i, w*, E, umax`; profiles: `u, v, theta, q, w*theta*`)
- Is it really an LES? (`E*, e, e*`)

Additional tasks:

- Calculate the relative humidity at initialization (use the Magnus formula…)
- Make a guess and start a new simulation with less humidity. How much humidity is needed for night-time dewfall?

Look up new variables in the PALM documentation!

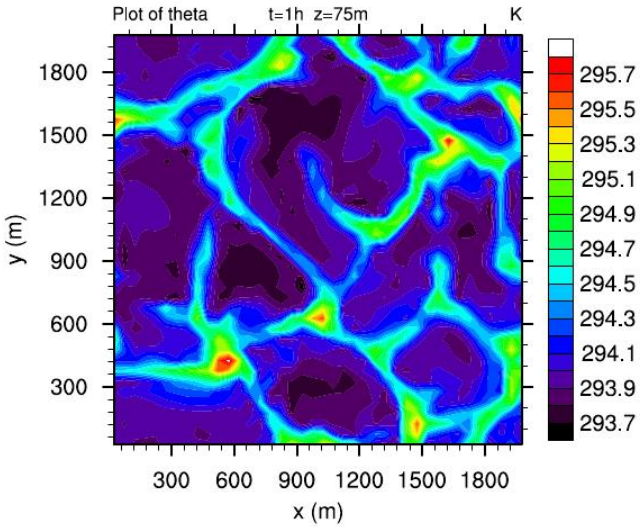
Be sure that you are aware of what you see and understand how the output is calculated!

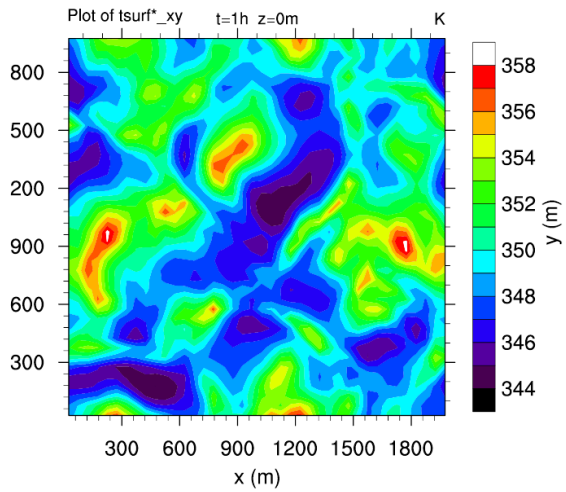
palmplot commands

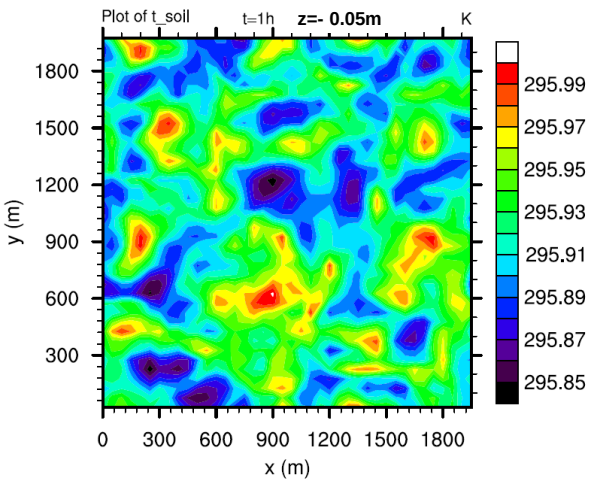
- time series:
```
palmplot ts file_1=e4_LSM2_ts.000.nc format_out=pdf \
  file_out=e4_LSM2_plot_ts var='qsws_liq rad_sw_in zi_theta E E*' no_rows=5
```
- profiles:
```
palmplot pr file_1=e4_LSM2_pr.000.nc format_out=pdf \
  file_out=e4_LSM2_plot_pr var='wu w*u* w”u” theta'
```

The palmplot-commands are just examples and may need to be adjusted to answer the questions!
Also the p3d-file may need to be edited to get appropriate outputs!

How does the surface and the soil react to the atmospheric turbulence?







Characteristics of a convective boundary layer:

- typical hexagonal cells
- higher temperature at edges
- lower temperature inside

Patterns at the surface:

- hexagonal cells
- higher temperature at edges
- **but additional temperature maximum in the centers**
- lower temperature in-between

In the soil:

- surface patterns persist in upper soil
- **in -5 cm surface patterns combine with others and become indistinguishable from oneanother**

What is the influence of a thinner/thicker first soil layer on the reaction of the surface temperature under atmospheric changes?

- It changes the storage term *S* and the ground heat flux *G*. The thinner the first soil layer, the smaller the heat capacity of the first layer and the greater the ground heat flux. A finer soil configuration can react faster to the atmosphere.

$$S = R_{net} - G - H - LE$$

$$S = \rho c_p dz_{soil,1} \frac{dT_0}{dt}$$

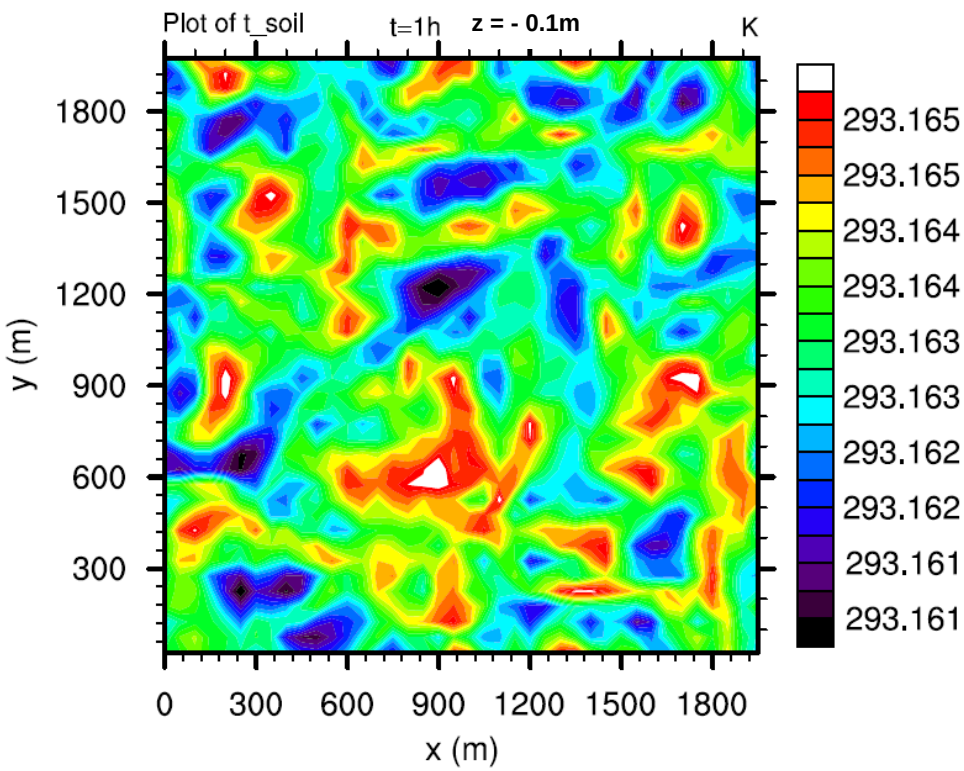
$$R_{net} = const.$$

$$G = \frac{\lambda}{dz_{soil,1}} \left(T_0 - T_{soil,1}\right)$$

$$H = \frac{\rho c_p}{r_a} (\theta_1 - \theta_0)$$

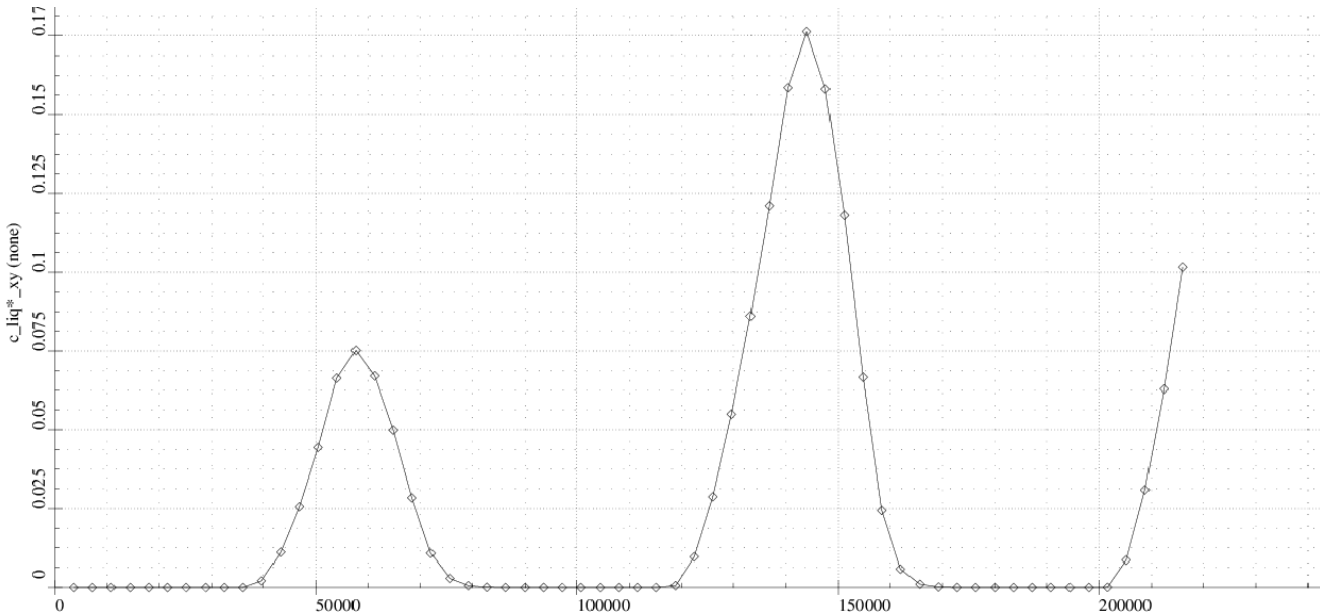
$$LE = 0$$

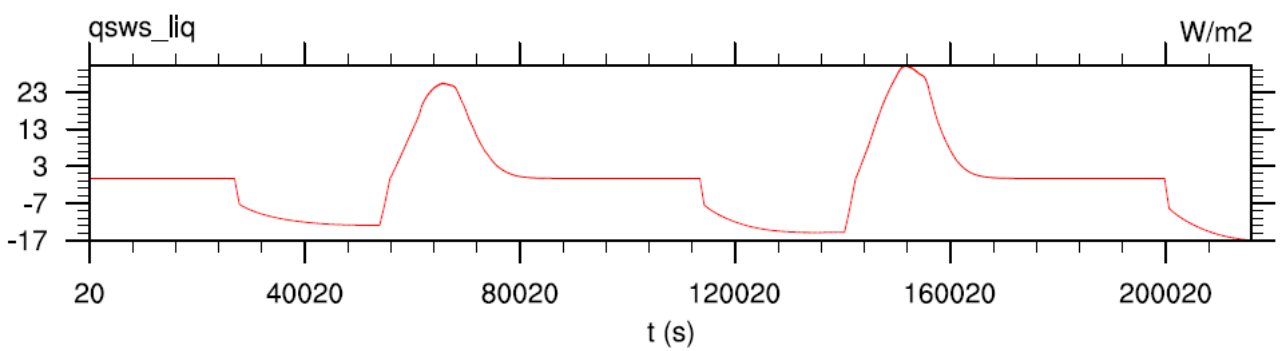
Down to which depth does the soil feel the surface forcing?



At 10 cm depth, the temperature variation is less than 0.005 K.

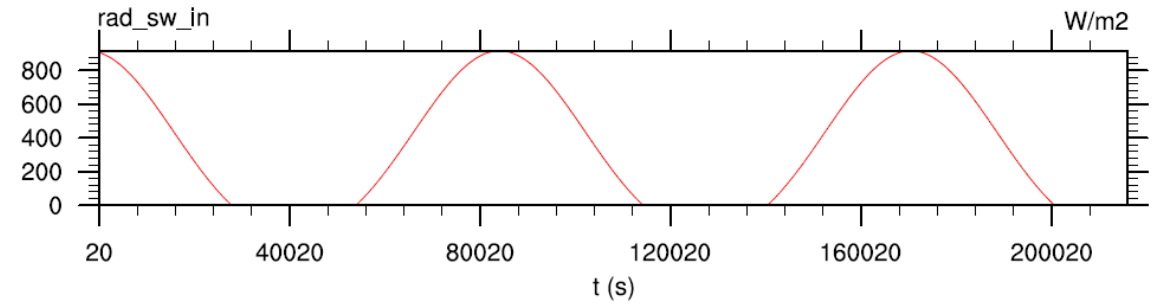
Can you observe dewfall? (time series: `qsws_liq`, `xy-section: c_liq`)

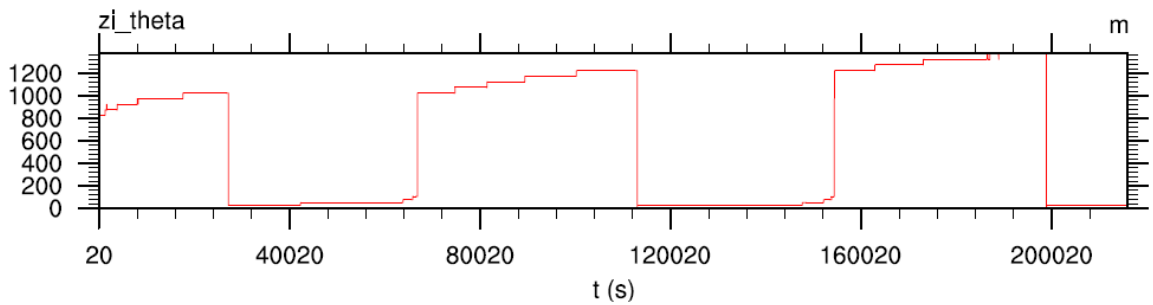


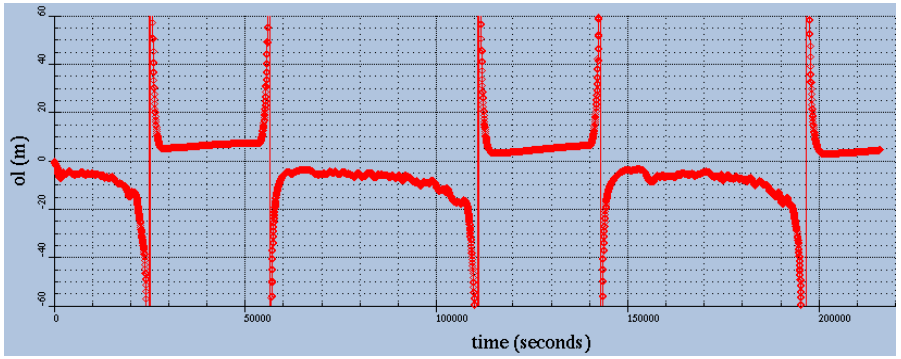


- Up to 10% of the vegetation is covered with dew.
- Dewfall sets in at 21:10 UTC (33000 s) and the vegetation is dry again at 10:20 UTC (80400 s) the next day.

What happens at night?
(time series: `ol, zi_theta, w*, E, umax`; profiles: `u, v, theta, q, w*theta*`)







- Boundary layer collapses at 19:30 UTC (27000 s), develops again at 7:30 UTC (70200 s) the next day.
- The Obukhov length shows that the ABL is unstable during day and stable at night.

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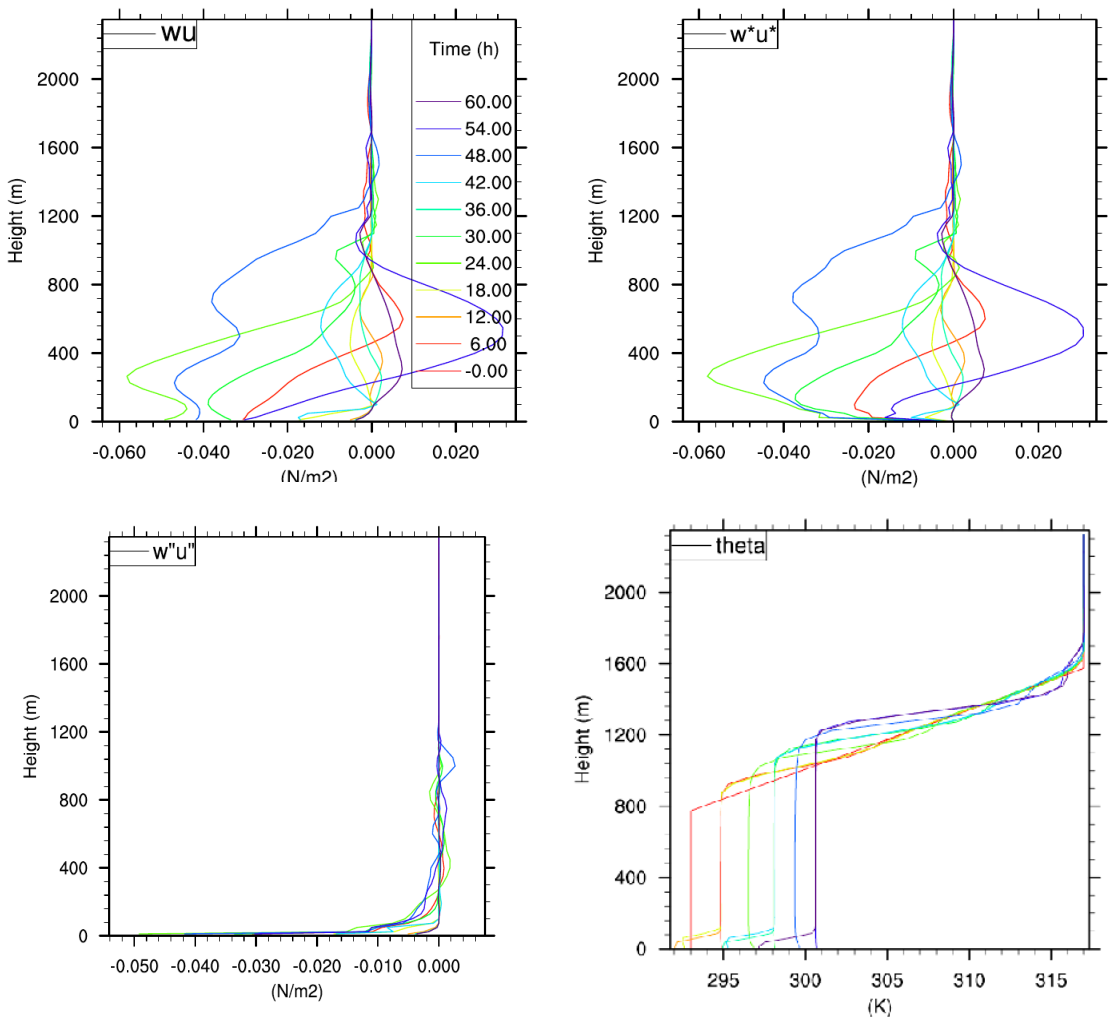
Part 2 – Answers



Is it really an LES?

- The nocturnal ABL is insufficiently resolved.
- Therefore, the simulation does not suffice the requirements of an LES **at night**.

- In the morning, convection develops and the LES results are reliable again (e.g., van Stratum, B. and B. Stevens, 2015).
- The analysis of night-time dewfall is not fully reliable. In reality, less or more dewfall could occur, if i.e. warmer or more humid air were mixed to the surface.
- Night-time ABLs are only resolved, if the LES uses a very small grid size (~1m)!

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Part 2 – Answers

Calculate the relative humidity at initalization (use the Magnus formula…)

- ca. 88%

Make a guess and start a new simulation with less humidity. How much humidity is needed for night-time dewfall?

- With a mixing ratio of 0.009 (relative humidity 61%), dewfall only covers less than 1% of the vegetation